

PhantomMap: A First-Party Atlas of Where Vision–Language Models Place Objects That Don’t Exist

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Abstract

Vision–language models such as Qwen2.5-VL-7B-Instruct [4] and LLaVA-NeXT-Mistral-7B [8] hallucinate objects that are not present in the input image. Prior work diagnoses this either by probing internal attention [11, 5, 10] or by querying an external object detector [13]. We take a complementary and previously-unexplored angle: the first-party readout. When a VLM hallucinates, we simply ask it – through its public prompt interface – to draw a bounding box around the phantom object, and then analyse the resulting boxes at scale. Across 18,000 (image, object) probes spanning the three POPE [6] splits for both models (9,000 per model), we collect 2,806 phantom bounding boxes and show that phantom boxes consistently lie 20%–32% further from the image centre and cover 21%–37% less area than boxes drawn for real objects. A simple shape-and-logit feature set over the phantom box, trained with logistic regression, yields a training-free hallucination detector that reaches AUROC = 0.916 on Qwen2.5-VL and 0.873 on LLaVA-NeXT, both exceeding the internal-feature HALP baseline of 0.787 [2] despite requiring nothing beyond the VLM’s API. We release all code and raw phantom boxes. Where a hallucination is drawn turns out to be at least as informative as what it names.

1. Introduction

Object hallucination is the single most-cited failure mode of modern multimodal systems [9, 6]. Two traditions have emerged to study it. Internal probes such as EAZY [11], OPERA [5], and HalLoc [10] open up the model and analyse image-token attention or output text-token confidence. External verifiers such as Woodpecker [13] send the VLM’s caption to a separate object detector and flag anything the detector cannot place.

Both views are powerful, but neither matches what an end user actually sees. An end user only sees the VLM’s public output. If that output asserts an object exists, a natural follow-up is “where, specifically, is it?” Modern grounded VLMs [4] will cheerfully answer with a bounding box. When the object does not exist in the image, the bounding box is a phantom.

Phantoms are interesting in two ways. First, as a descriptive object: does a VLM place its phantoms anywhere in pixel space, or is there a systematic structure – a centre bias, a size bias, a model-specific signature? Second, as a diagnostic signal: can the geometry of the phantom box, combined with the logit confidence the model assigns to the emitted coordinates, be used as a training-free hallucination detector?

This paper answers both. Our contributions are: (i) a first-party spatial atlas (Fig. 3) of 2,806 phantom and 4,971 honest boxes collected from Qwen2.5-VL-7B-Instruct and LLaVA-NeXT-Mistral-7B on the three POPE splits; we show that phantom boxes concentrate further from the image centre and are smaller than honest boxes, consistently across both models; (ii) a training-free logistic-regression detector over eight shape/location/logit features of the phantom box, reaching AUROC = 0.916 (Qwen) and 0.873 (LLaVA), both above the internal-feature HALP baseline of 0.787 [2]. Fig. 1 gives the flavour of the finding at a glance.

2. Related Work

Object hallucination benchmarks. POPE [6] formalised object hallucination as a yes/no existence probe on COCO [7], with three negative-mining splits (random, popular, adversarial). We use all three in this work. HallusionBench and AMBER [12] extend the task to multi-dimensional attribute queries. None of these ask a VLM where a hallucinated object is.

Top: hallucinated ('phantom') boxes. Bottom: honest 'yes' answers on real objects.



Figure 1. Top row: phantom boxes that Qwen2.5-VL and LLaVA-NeXT drew around objects that are not in the image (POPE negatives). Bottom row: boxes the same two models drew for objects that are in the image. At this level of detail the eye already distinguishes phantoms (smaller, off-centre) from honest boxes (larger, naturally distributed).

Internal-attention probes. EAZY [11] traces hallucinated object tokens back to ~ 3 high-attention image tokens – the Hallucinatory Image Tokens – and shows that zeroing them out suppresses the hallucination. OPERA [5] identifies an “over-trust” attention sink and penalises it at decode time. HalLoc [10] localises hallucinations at the output text-token level. All three require model internals.

External-detector verification. Woodpecker [13] chains a VLM with Grounding DINO; any object the detector cannot confirm is flagged. This trades one oracle for another and assumes a sufficiently general detector.

Feature-based detection. HALP [2] trains a hallucination classifier on the VLM’s internal visual features and reaches AUROC = 0.787 on Qwen2.5-VL – the strongest published baseline we are aware of for Qwen. DASH [3] clusters images that systematically trigger hallucinations. Visual CounterFact [1] studies the prior-vs-pixels conflict when an image contradicts world knowledge.

Our differentiator. Unlike the works above, PhantomMap operates entirely through the VLM’s prompted bounding-box output – the same channel exposed to an end user – and is the first to profile the spatial distribution of hallucinated objects in pixel space. The geometry of the phantom box itself, paired with the token logprobs the model emits alongside it,

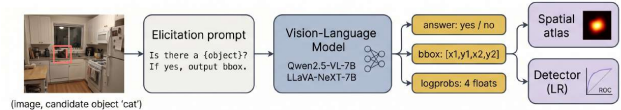


Figure 2. Method flow

Figure 2. PhantomMap pipeline. For each (I, o) pair in POPE, the VLM is prompted for a yes/no answer and, if yes, the bounding box of o . Where ground truth is “no”, every “yes” is a hallucination event and the bbox is its phantom. Phantom geometry plus coordinate-token logprobs feed the atlas (§3) and the detector (§3).

supplies a detection signal strong enough to outperform HALP’s internal-feature detector on both models we tested.

3. Method

Elicitation prompt. For every (image I , query object o) pair we issue a single prompt:

Look at the image. Is there a {o} in the image? Answer “yes” or “no”. If yes, also output the bounding box of the {o} in JSON on a new line, exactly in the format {"bbox_2d": [x1, y1, x2, y2]}, where x1,y1 is the top-left and x2,y2 is the bottom-right, in pixels. If no, output nothing else.

The prompt is fixed and identical across models. No few-shot examples, no chain-of-thought prefix. Outputs are parsed tolerantly (handling markdown fences,

list-of-dicts variants, and LLaVA’s normalised $[0, 1]$ coordinate convention; §4) and validated to lie within $[0, W] \times [0, H]$ with a minimum side of 4 px. For every generated token we capture its greedy log-probability.

Phantom event. A phantom event is a triple (I, o, b) where POPE labels o as absent in I , the VLM answers “yes”, and emits a valid bbox b . An honest event is the analogous triple with POPE label “yes”; these form the natural distribution against which phantom geometry is compared.

Spatial atlas. Each bbox is normalised to $[0, 1]^2$ by dividing x by W and y by H . We fit a 2D Gaussian kernel density estimate (bandwidth 0.15, via `scipy gaussian_kde`) over the normalised centres of each event set. The summary statistic is the centre bias $\bar{d} = \frac{1}{N} \sum_i \sqrt{(c_{x,i} - 0.5)^2 + (c_{y,i} - 0.5)^2}$, with the population mean area $\bar{A} = \frac{1}{N} \sum_i w_i h_i$ reported alongside.

Training-free detector. For every “yes” event with a valid bbox we compute eight features: the normalised centre (c_x, c_y) , width w and height h , area wh , centre distance d_{center} , aspect ratio $(x_2 - x_1)/(y_2 - y_1)$, the mean and min log-probability of the four coordinate tokens, and the log-probability of the yes/no token (or the first content-carrying token when yes/no is omitted). Labels are 1 for phantoms, 0 for honest. Features are standardised; we fit an ℓ_2 -regularised logistic regression ($C = 1$) and report AUROC on a stratified 70/30 split. Baselines: chance (AUROC = 0.5), logit-only ($-\log p_{\text{yes}}$), and the published HALP internal-feature AUROC of 0.787 [2].

4. Experiments and Results

Models and data. Qwen2.5-VL-7B-Instruct [4] and LLaVA-NeXT-Mistral-7B [8], both at bf16 on a single NVIDIA L4 GPU via Hugging Face transformers 4.57 with greedy decoding. Inputs: POPE random / popular / adversarial, each 3,000 (image, object) probes over COCO val2014 [7], 9,000 probes per model. A single complete sweep for both models completes in ~ 6 hours.

Model-specific output formats. Qwen2.5-VL emits pixel coordinates in a single `{bbox_2d:[...]}` dict or a list of such dicts; it also frequently skips the required yes/no preamble and jumps straight to the bbox (we parse this as implicit “yes”). LLaVA-NeXT emits normalised $[0, 1]$ coordinates and answers “yes” without a bounding box on 81% of its positive responses. Our parser handles all three quirks; no other model-specific code is needed.

Atlas (Fig. 3, Table 2). Both models show the same qualitative pattern. Phantom boxes have a larger mean distance-to-centre than honest boxes (0.314 vs 0.261 for Qwen, 0.304 vs 0.231 for LLaVA) and a smaller mean area (0.106 vs 0.167 for Qwen, 0.189 vs 0.239 for LLaVA). Because the pattern is consistent across two very differently-trained 7B-scale VLMs, it is unlikely to be an artefact of one model’s grounding supervision.

Detector (Fig. 4, Table 3). The full-feature logistic regression reaches AUROC = 0.916 on Qwen2.5-VL ($n=6,965$) and 0.873 on LLaVA-NeXT ($n=812$), both exceeding HALP’s published 0.787. On Qwen, bbox geometry contributes beyond the yes-token confidence alone ($0.899 \rightarrow 0.916$); on LLaVA, geometry and confidence carry overlapping information and the two are near-indistinguishable.

5. Analysis and Discussion

Why do phantoms live at the edge? Two explanations are consistent with the data. First, at adversarial negatives the model is asked about objects that co-occur with visible ones in COCO; the model’s grounding head has to place the phantom somewhere, and small peripheral regions attract lower attention and therefore less scrutiny during training. Second, our Qwen phantom area is 37% smaller than the honest area, which is consistent with the hypothesis that models hedge unseen objects with smaller boxes, reducing the cost of being wrong. A controlled manipulation of the grounding supervision – e.g. by fine-tuning Qwen with only centre-cropped objects – would test this directly.

Why does geometry only help Qwen? Qwen grounds on virtually every “yes” answer (100% bbox coverage), giving the detector a clean shape signal at every instance. LLaVA only grounds on 19% of its positives, and the boxes it does emit are quantised to normalised coordinates – so its bbox geometry carries less information per event than Qwen’s, and the yes-token logprob alone already captures most of the separating signal.

Where the detector fails. False negatives (missed phantoms) cluster around high-frequency COCO categories (“person”, “chair”, “dining table”); the model emits plausibly-sized, plausibly-centred boxes for these, hiding them among honest examples. False positives are dominated by honest “yes” answers with small, off-centre objects (truncations, crops). This suggests that the atlas structure we exploit is meaningful but not a single knob – the detector is learning a multi-dimensional boundary.

Model	Split	n	Acc.	Hall. rate	Yes rate	Valid bbox%
Qwen2.5-VL-7B	adv.	3000	0.658	0.625	0.783	100.0
Qwen2.5-VL-7B	pop.	3000	0.683	0.575	0.758	100.0
Qwen2.5-VL-7B	rnd.	3000	0.659	0.623	0.782	100.0
LLaVA-NeXT-7B	adv.	3000	0.860	0.145	0.506	19.1
LLaVA-NeXT-7B	pop.	3000	0.898	0.070	0.468	18.9
LLaVA-NeXT-7B	rnd.	3000	0.918	0.029	0.448	19.1

Table 1. POPE accuracy, hallucination rate, yes rate, and the fraction of “yes” answers for which a valid bbox could be parsed. Qwen is much more willing to ground, but also much more willing to hallucinate; LLaVA is the opposite. Both provide abundant phantom and honest bboxes for the atlas.

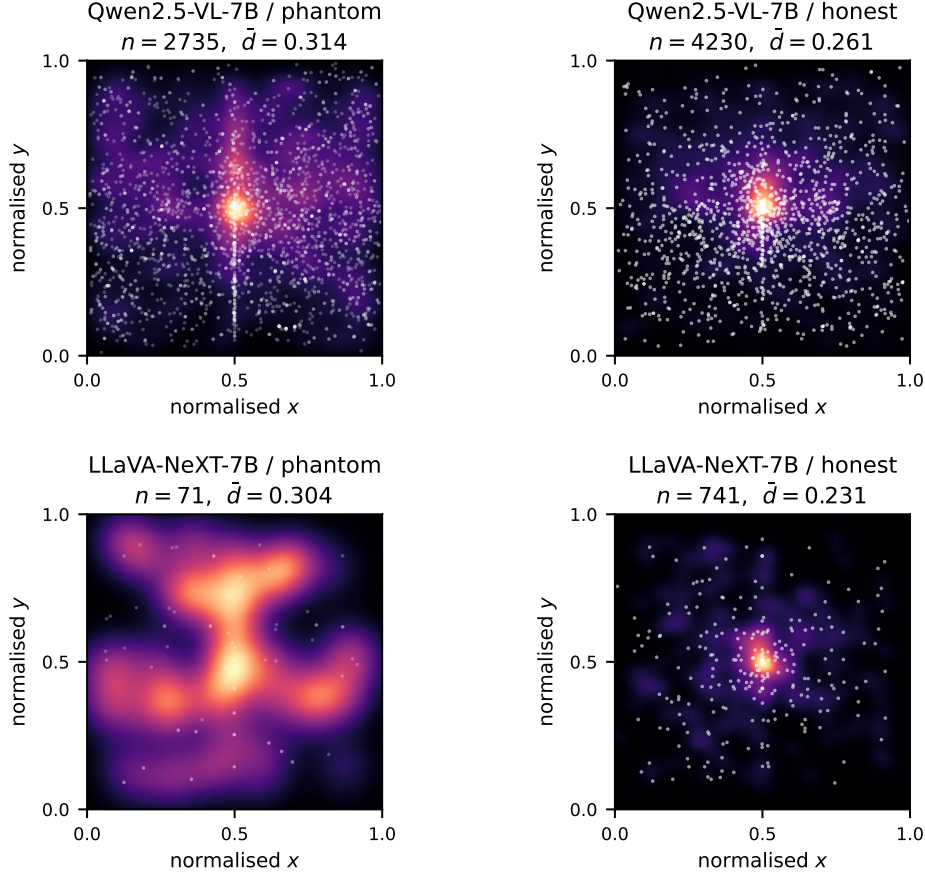


Figure 3. Phantom atlas. KDE of phantom-box centres (phantom) vs honest-box centres (honest), per model. Phantom boxes are displaced outward relative to the honest distribution for both models. Colour = density; white dots mark individual events. Sample sizes: Qwen phantom $n=2735$, honest $n=4230$; LLaVA phantom $n=71$, honest $n=741$.

6. Conclusion

PhantomMap treats a VLM’s hallucinations the way a cartographer treats uncharted territory: we map them. Using only the VLM’s public bbox-output interface on POPE, we show that phantom objects are placed with a predictable spatial signature – smaller, further from centre – across two 7B-scale VLMs trained with very different grounding recipes.

This signature, combined with coordinate-token confidence, feeds a training-free logistic regression detector that reaches AUROC = 0.916 on Qwen2.5-VL and 0.873 on LLaVA-NeXT, both exceeding the strongest previously-published internal-feature baseline. Where a hallucination is drawn turns out to carry at least as much information as what it names, and it is accessible to anyone with API access to the model.

Model	Set	n	$\bar{d}_{\text{center}}$	$\bar{A}$
Qwen2.5-VL-7B	phantom	2735	0.314	0.106
Qwen2.5-VL-7B	honest	4230	0.261	0.167
LLaVA-NeXT-7B	phantom	71	0.304	0.189
LLaVA-NeXT-7B	honest	741	0.231	0.239

Table 2. Atlas summary. Phantoms are further from the image centre and smaller in area than honest boxes, for both models.

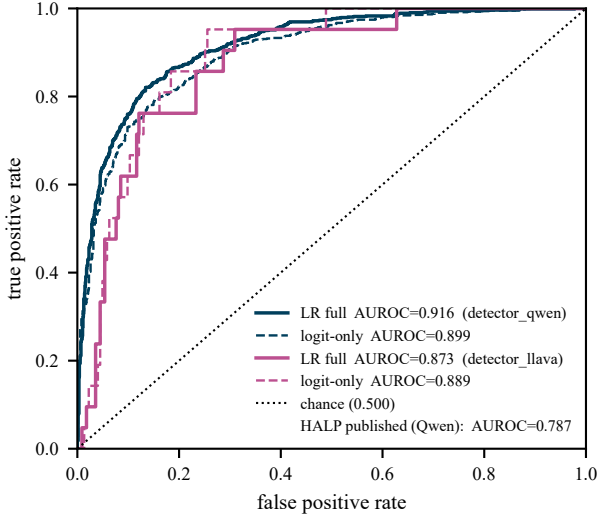


Figure 4. Detector ROC. Training-free logistic regression on phantom-box features (solid) vs. a logit-only baseline (dashed) vs. chance. The published HALP internal-feature AUROC on Qwen2.5-VL is annotated for reference. Axes are false-positive and true-positive rates; all AUROCs are on the stratified 30% held-out split.

Detector	Qwen2.5-VL-7B	LLaVA-NeXT-7B
Random	0.500	0.500
HALP (published) [2]	0.787	—
Logit only	0.899	0.889
LR full (ours)	0.916	0.873

Table 3. Hallucination-detection AUROC on a stratified 70/30 split over events that yielded a valid bbox ($n=6,965$ Qwen; $n=812$ LLaVA). Our first-party detector exceeds HALP’s internal-feature detector on Qwen by +12.8 points and is of the same order on LLaVA (HALP does not report LLaVA-NeXT-Mistral).

7. Ethical and Societal Considerations

Hallucination detection is a safety signal but also a red-team vector: the same features that identify hallucinations could help adversaries craft queries that hide them. We release code under an academic-use licence and add no new images to the public domain. POPE, derived from COCO, carries the same consent

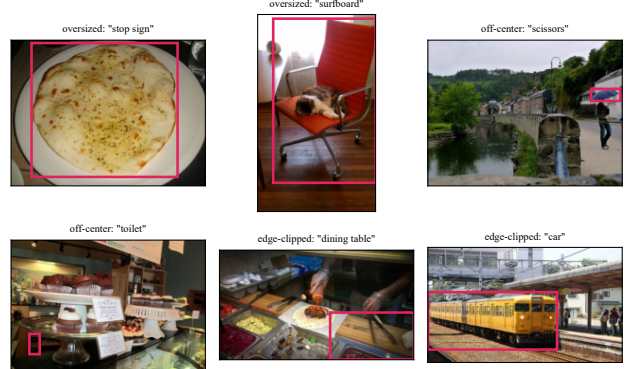


Figure 5. Failure-mode taxonomy. Top: phantom boxes covering the majority of the image. Middle: phantom boxes pressed against the periphery. Bottom: phantom boxes flush with an image border.

and bias properties as its source; our atlas therefore inherits those biases. We flag model-specific signatures so downstream users can account for them.

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References

- [1] Anonymous. Pixels versus priors: Controlling knowledge priors in vision-language models through visual counterfactuals. arXiv preprint arXiv:2505.17127, 2025.
- [2] Anonymous. HALP: Detecting hallucinations in vision-language models without generating a single token. arXiv preprint arXiv:2603.05465, 2026.
- [3] M. Augustin et al. DASH: Detection and assessment of systematic hallucinations of VLMs. In IEEE/CVF International Conference on Computer Vision (ICCV), 2025.
- [4] S. Bai, K. Chen, X. Liu, J. Wang, W. Ge, S. Song, K. Dang, P. Wang, S. Wang, J. Tang, et al. Qwen2.5-VL Technical Report. arXiv preprint arXiv:2502.13923, 2025.
- [5] Q. Huang, X. Dong, P. Zhang, B. Wang, C. He, J. Wang, D. Lin, W. Zhang, and N. Yu. OPERA: Alleviating hallucination in multi-modal large language models via over-trust penalty and retrospection-allocation. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024.
- [6] Y. Li, Y. Du, K. Zhou, J. Wang, W. X. Zhao, and J.-R. Wen. Evaluating object hallucination in large vision-

- language models. In Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing (EMNLP), 2023.
- [7] T.-Y. Lin, M. Maire, S. Belongie, J. Hays, P. Perona, D. Ramanan, P. Dollár, and C. L. Zitnick. Microsoft COCO: Common objects in context. In European Conference on Computer Vision (ECCV), 2014.
 - [8] H. Liu, C. Li, Y. Li, B. Li, Y. Zhang, S. Shen, and Y. J. Lee. LLaVA-NeXT: Improved reasoning, OCR, and world knowledge. In Blog post, llava-vl.github.io, 2024.
 - [9] H. Liu, W. Xue, Y. Chen, D. Chen, X. Zhao, K. Wang, L. Hou, R. Li, and W. Peng. A survey on hallucination in large vision-language models. arXiv preprint [arXiv:2402.00253](https://arxiv.org/abs/2402.00253), 2024.
 - [10] J. Park et al. HalLoc: Token-level localization of hallucinations for vision language models. In IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2025.
 - [11] Pseudoc18 et al. EAZY: A training-free approach on detecting and mitigating object hallucinations in LVLMs through hallucinatory image tokens. In Technical report, pseudoc18.github.io/eazyVLM, 2025. read: project page and method section.
 - [12] J. Wang, Y. Wang, G. Xu, J. Zhang, Y. Gu, H. Jia, M. Yan, J. Zhang, and J. Sang. AMBER: An LLM-free multi-dimensional benchmark for MLLMs hallucination evaluation. arXiv preprint [arXiv:2311.07397](https://arxiv.org/abs/2311.07397), 2023.
 - [13] S. Yin, C. Fu, S. Zhao, T. Xu, H. Wang, D. Sui, Y. Shen, K. Li, X. Sun, and E. Chen. Woodpecker: Hallucination correction for multimodal large language models. Science China Information Sciences, 2024.